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General Comment

Please find attached the comments of the Wireless Infrastructure Association ("WIA")

Attachments

WIA Comments in AI Action Plan (03.15.25)

**Before the
NETWORKING AND INFORMATION TECHNOLOGY RESEARCH AND
DEVELOPMENT NATIONAL COORDINATION OFFICE,
NATIONAL SCIENCE FOUNDATION
Alexandria, VA 22314**

In the Matter of

*Request for Information on the
Development of an Artificial Intelligence
Action Plan*

NSF-0001-3479

COMMENTS OF THE WIRELESS INFRASTRUCTURE ASSOCIATION¹

The Wireless Infrastructure Association (“WIA”)² submits these comments in response to the National Science Foundation’s (“NSF”) Request for information on developing an Action Plan to sustain and enhance America’s Artificial Intelligence (“AI”) landscape.³ WIA appreciates the Administration’s support of the critical AI ecosystem and this opportunity to provide the unique insights of the wireless infrastructure industry in development of the Action Plan. Securing and maintaining the United States’ position as a leader in AI will require a holistic approach that considers the needed inputs at the physical and information layers along with support of the outputs to ensure new AI tools can be effectively leveraged. These comments focus on the importance of supporting and maintaining the needed infrastructure to enable advancements in the AI landscape,

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² WIA represents the businesses that build, develop, own, and operate the nation’s wireless infrastructure. WIA advocates for the widespread, responsible deployment of wireless infrastructure to enable connectivity everywhere.

³ *Development of an Artificial Intelligence (AI) Action Plan*, 90 Fed. Reg. 9088, Request for Information, NAT’L. SCIENCE FOUNDATION (Feb. 6, 2025) [*AI RFI*].

including supporting the deployment of compute power—both at the network core and edge—along with improving access to fixed and mobile broadband.

It is paramount that the Administration consider the infrastructure needs of current and future AI applications. While significant attention is being paid to the various software and models being used to develop these tools, these advancements are ultimately enabled by the physical infrastructure; particularly data centers as well as communications networks. In order to achieve a robust and successful AI ecosystem, the Administration should consider how its policies can support infrastructure deployment. In the immediate term, the Administration should encourage Congress to act to enable infrastructure deployment by reinstating the Federal Communications Commission’s (“FCC”) authority to conduct spectrum auctions and enact meaningful permitting reform as contemplated last Congress in the American Broadband Deployment Act.⁴

I. Physical Infrastructure Plays a Critical Role in the AI Ecosystem.

AI applications promise to be transformative for nearly every industry, but to ensure the widespread use and availability of these applications there needs to also be sufficient infrastructure. For example, data centers are needed to provide sufficient computing power and data storage for training and maintaining models, while fixed and mobile broadband networks enable access and functionality for these tools. Accordingly, the Action Plan being developed should be sure to promote policies that ensure the predictable, proportionate, and timely deployment of this infrastructure.

A. Various Data Center Deployments are Needed to Realize the Potential of AI.

Data centers of all types will be necessary to train, support, and utilize AI models and tools. Particularly in our increasingly interconnected world, on demand access to data and computing

⁴ H.R. 3557, 118th Cong. (2023).

power are key considerations for businesses and consumers.⁵ Data centers are not, however, monolithic. Generally, they are divided into three classes based on their size and application. The largest centers, called hyperscalers, are usually geared towards a single application such as supporting large digital platforms or training and developing new data applications like AI/ML models.⁶ The next tier can also be large in size but is operated as a neutral host, leasing out space and resources to multiple customers.⁷ These colocation data centers allow operations to utilize cloud and virtualized services without incurring the high cost of building and operating a data center, creating economic and environmental efficiencies. Colocation data centers house the computing infrastructure for a wide range of America's business and government interests and allow valuable interconnection between tenants providing secure and low-latency data transfers.⁸ The next tier, edge data centers, are usually smaller in form but are still impactful for consumers, adding needed capacity and compute power to communications networks that are critical for end user enterprises and consumers to take full advantage of advanced capabilities.⁹

⁵ *Economic, Environmental, and Social Impacts of Data Centers in the US*, PWC at 5, 10 (Sept. 2024), <https://static1.squarespace.com/static/63a4849eab1c756a1d3e97b1/t/65037be19e1dbf4493d54c6e/1694727143662/DCC-PwC+Impact+Study.pdf> (examining the requirements of businesses to have access to data-intensive services that will rely on data center infrastructure).

⁶ See *Data Center RFC* at 71890. See also *Economic, Environmental, and Social Impacts of Data Centers in the United States* at 12 (defining hyperscale data centers as having at least 5,000 servers over 10,000 square feet and offering at least 40MW of power capacity).

⁷ *Ecosystems and Edge: The Data Center as a Transactional Marketplace*, DATA CENTER FRONTIER (Sept. 20, 2021), <https://www.datacenterfrontier.com/voices-of-the-industry/article/11427945/ecosystems-and-edge-the-data-center-as-a-transactional-marketplace> (examining the rising interest in data centers that can provide on premise interconnection).

⁸ See *Understanding Data Center Interconnection* (providing examples and benefits of different kinds of data center interconnection). See also *Colocation*, CROWN CASTLE, <https://www.crowncastle.com/resources/solution-sheet/colocation.pdf> (last visited Nov. 4, 2024) (providing benefits and offerings in colocation data centers).

⁹ See, e.g. *Edge Data Centers: Riding the 5G and IoT Wave* at 3-4; *Extending Clouds to the Edge: The Case for Aggregation Edge Data Centers*, AMERICAN TOWER (July 8, 2024), <https://www.americantower.com/us/media-hub/extending-clouds-edge-case-aggregation-edge-data-centers>; Hillery Hunter and Ed Knapp, *American Tower and*

While hyperscale centers have demonstrated immense value in training AI models, colocation and edge data centers will become increasingly important in the AI landscape as developments move towards reference models and consumers expect access to these tools wherever they go.¹⁰ Several next generation AI applications will require improved data processing and storage at the edge including autonomous vehicles and AR/VR.¹¹ In order to meet these expectations service providers will need to continually improve their networks, including by moving resources closer to the end-user. WIA encourages the administration to promote policies in the Action Plan that will support all types of data centers to enable a host of applications.

B. The Administration Should Support Broadband Connectivity to Support AI Development and Adoption

Similarly, access to high-speed broadband—over both fixed and wireless networks—will be table stakes for participating in the AI revolution. However, millions of Americans still live in areas that lack sufficient broadband access or capacity to fully utilize these tools.¹² Smart policies

IBM to Bring Edge Cloud Services to Enterprises for Enhanced Innovation and Customer Experiences, IBM (Jan. 5, 2024), <https://newsroom.ibm.com/American-Tower-and-IBM-to-Bring-Edge-Cloud-Services-to-Enterprises-for-Enhanced-Innovation-and-Customer-Experiences>; *Edge Computing*, SBA EDGE, <https://www.sbaedge.com/edge-computing/> (last visited Nov. 4, 2024) (“[Edge Computing] enables organizations to run workloads with very low latency, offering a new approach to computing that eliminates the need for internet transit from remote locations to the data center.”).

¹⁰ See, e.g., *Extending Clouds to the Edge: The Case for Aggregation Edge Data Centers* (providing that demand for new technologies including hybrid cloud computing, AI, and 5G networks has required industry to seek data management and processing solutions at the edge); Dan Jones, *AI Workloads 5-Year Plan*, FIERCE NETWORK (Jun. 25, 2024), <https://www.fierce-network.com/cloud/ai-workloads-5-year-plan> (examining how mature applications of AI will require improved edge infrastructure).

¹¹ See *Edge Data Centers: Riding the 5G and IoT Wave* at 3-4 (providing examples of new technologies that will rely on edge data centers).

¹² *2024 Section 706 Report*, GN Docket No. 22-270, FED. COMM’NS. COMM’N. (rel. March 18, 2024) (finding approximately 7% of Americans lack access to broadband with a significant portion of those unconnected in rural areas).

and targeted federal support can help bridge the gap to bring fixed and mobile broadband to all communities and deliver the benefits of new technologies.

Even where connectivity exists, networks must be continually improved to meet growing consumer demand. Broadband traffic has steadily increased for many years and the introduction of AI tools is expected to provide a substantial increase in demand.¹³ Ubiquitous mobile broadband that can support data-intensive applications will also increasingly be important.¹⁴ Many exciting developments and applications of AI—such as realizing new efficiencies from smart sensors and the Internet of Things (“IOT”) to autonomous vehicles on the road and in the sky—will rely on the high-throughput low-latency mobile networks. In order to meet this demand network providers are continually deploying new infrastructure and improving existing deployments.

To support these efforts, Congress and the Administration must act. The FCC has been without crucial auction authority for over two years now which Congress must act to restore. Spectrum auctions have raised billions of dollars for the U.S. government, spur innovation, and enable new services over wireless networks. Reinstating this authority and identifying a pipeline of spectrum to be made available for commercial uses should be a priority for the current Congress.

¹³ *Ericsson Mobility Report, Q4 Update* (Feb. 2025), <https://www.ericsson.com/492af1/assets/local/reports-papers/mobility-report/documents/2024/ericsson-mobility-report-q4-2024-update.pdf> (providing ongoing mobile traffic demands and projected increases due to AI activity).

¹⁴ *AI Workloads 5-Year Plan*, (“Over the next five years, [AI] workloads will start to shift from high-end large training systems run by cloud hyperscalers and in data centers to the network edge”); *Edge Data Centers: Riding the 5G and IoT Wave*, PWC at 2 (July 2019), <https://www.pwc.com/us/en/industries/capital-projects-infrastructure/library/assets/pwc-edge-data-centers.pdf> (“The potential of smaller, locally located data centers to reduce latency, overcome intermittent connections, and store and compute data close to the end user make them ideal for many new, low-latency, data-heavy technologies. They may enable all-new business models, based on decentralization and speed.”).

Additionally, Congress has the opportunity to enact common-sense permitting reform as proposed in the American Broadband Deployment Act (“ABDA”).¹⁵ During the last session, Congress considered this permitting legislation that would codify key siting reforms adopted by the FCC over the past two decades. Having Congress adopt an effective permitting framework is critical to ensure a consistent and certain deployment landscape across the country. By taking these simple steps, Congress can expedite AI deployment and development while bringing connected benefits to millions of Americans.

II. Conclusion

The AI ecosystem is a vast landscape that relies on—and will significantly impact—a myriad of industries. Hardware, software, infrastructure, and service providers all have important roles in the development and implementation of AI technologies. In crafting the Action Plan, WIA encourages the Administration to consider the breadth of industries that will contribute to maintaining America’s place at the forefront of innovation. In particular, policies that incentivize investment and deployment of infrastructure today will have outsized benefits in ensuring ubiquitous access to these tools and spur innovation.

Respectfully submitted,

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¹⁵ H.R. 3557, 118th Cong. (2023).